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TOPIC 9

CALCULATING STARTUP CAPITAL REQUIREMENTS & MANAGING STARTUP, FIXED AND VARIABLE COSTS

Topic Objectives

1. Discuss how entrepreneurs acquire the resources they need to launch a business
2. Identify typical startup financial metrics
3. Explain the role of financial assumptions
4. Discuss how entrepreneurs should assess risk when calculating financial needs

Calculating Capital Startup Requirements

- ❑ The bottom line for any new venture is cash
 - ❑ Cash pays the bills, enables the firm to grow
- ❑ 1st, At feasibility analysis stage, need a financial picture of the firm supported by real data
- ❑ 2^d, Investors will confirm that most important are the assumptions behind the numbers
- ❑ 3rd, no set of financial projections survives first contact with customers

Identifying Startup Resource Requirements

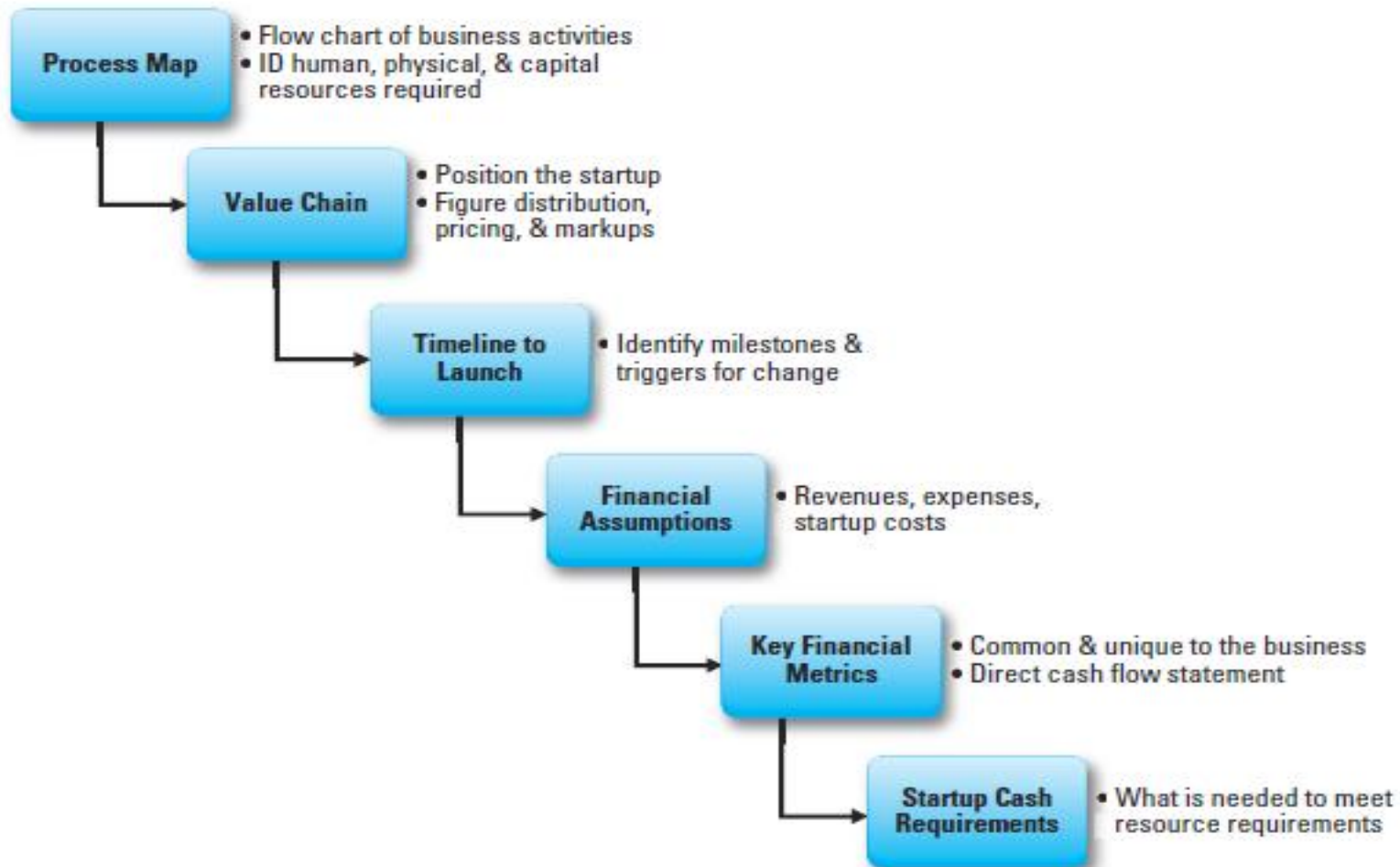
❑ Startup resources include:

- People (founding team, employees, advisors, etc.)
- Physical assets (equipment, inventory, office, etc.)
- Financial resources (cash, equity, debt)
- Intellectual resources (brand, patents, etc.)

❑ Decisions about resources relate to:

- 1. Value proposition or solution
 - 2. Distribution channels
 - 3. Customer relationships
 - 4. Revenue generation
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Figure 9.1- Steps for Calculating Startup Capital Requirements



9.1a Construct a Business Process Map

- ❑ Recall four categories of resources: human, intellectual, financial, physical
- ❑ To identify accurately the resources required must know exactly how the business works
- ❑ Create a process map that details how information flows through the business
 - Take a virtual tour of your business during a single day, listing all functions, activities, people, equipment, supplies and space required

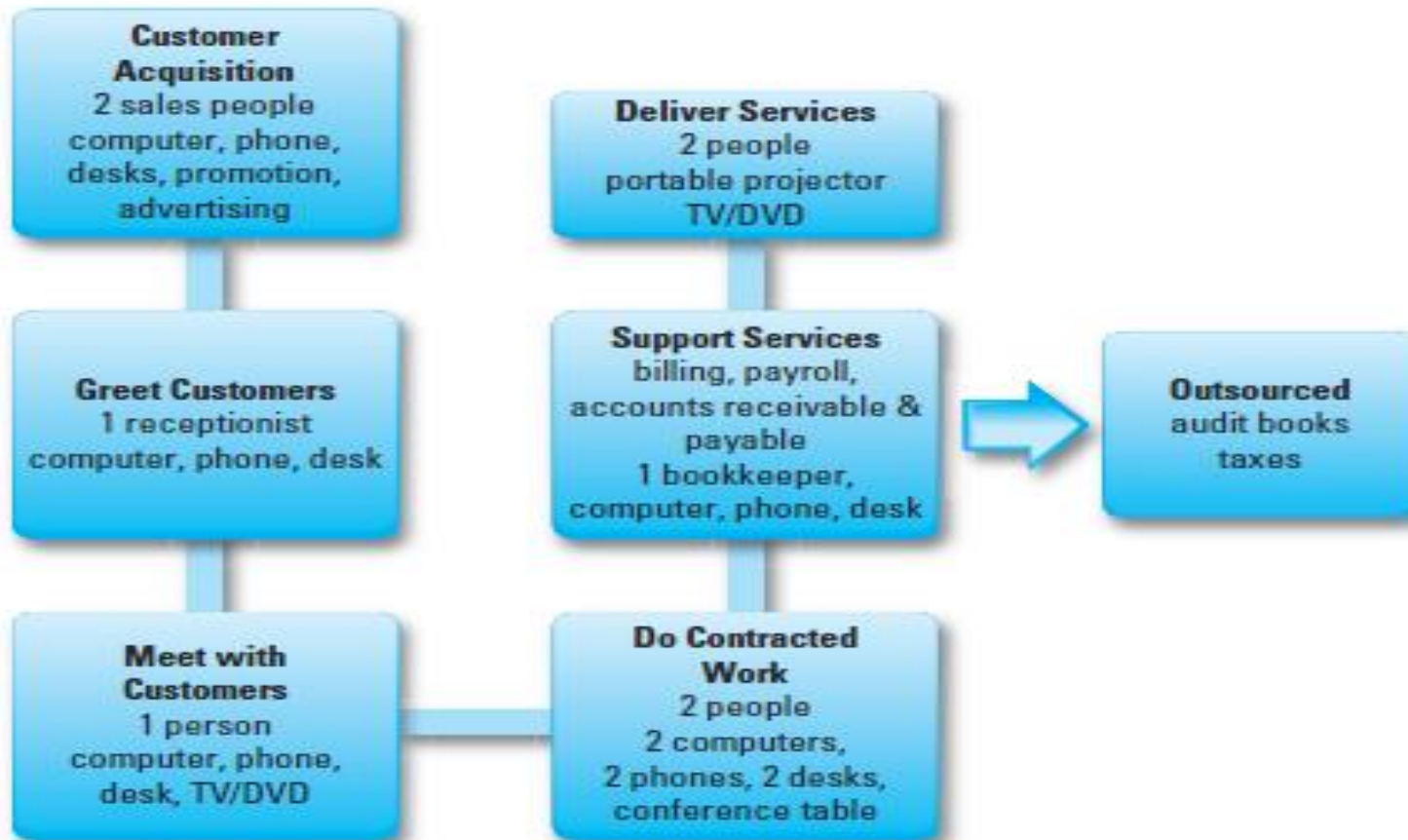
9.1a Construct a Business Process Map

□ Begin at the front door and ask:

- 1. Who does the work in this business?
- 2. Where do these people work?
- 3. What do they need to do the work (equipment, supplies, space, etc.)?
- 4. What information is being generated (work orders, invoices, customer lists, etc.)?
- 5. Where does that information go?

□ Then make lists of what's needed to complete a certain process or activity

Figure 9.2- A Virtual Tour of a Service Business



9.1b Position the Venture in the Value Chain

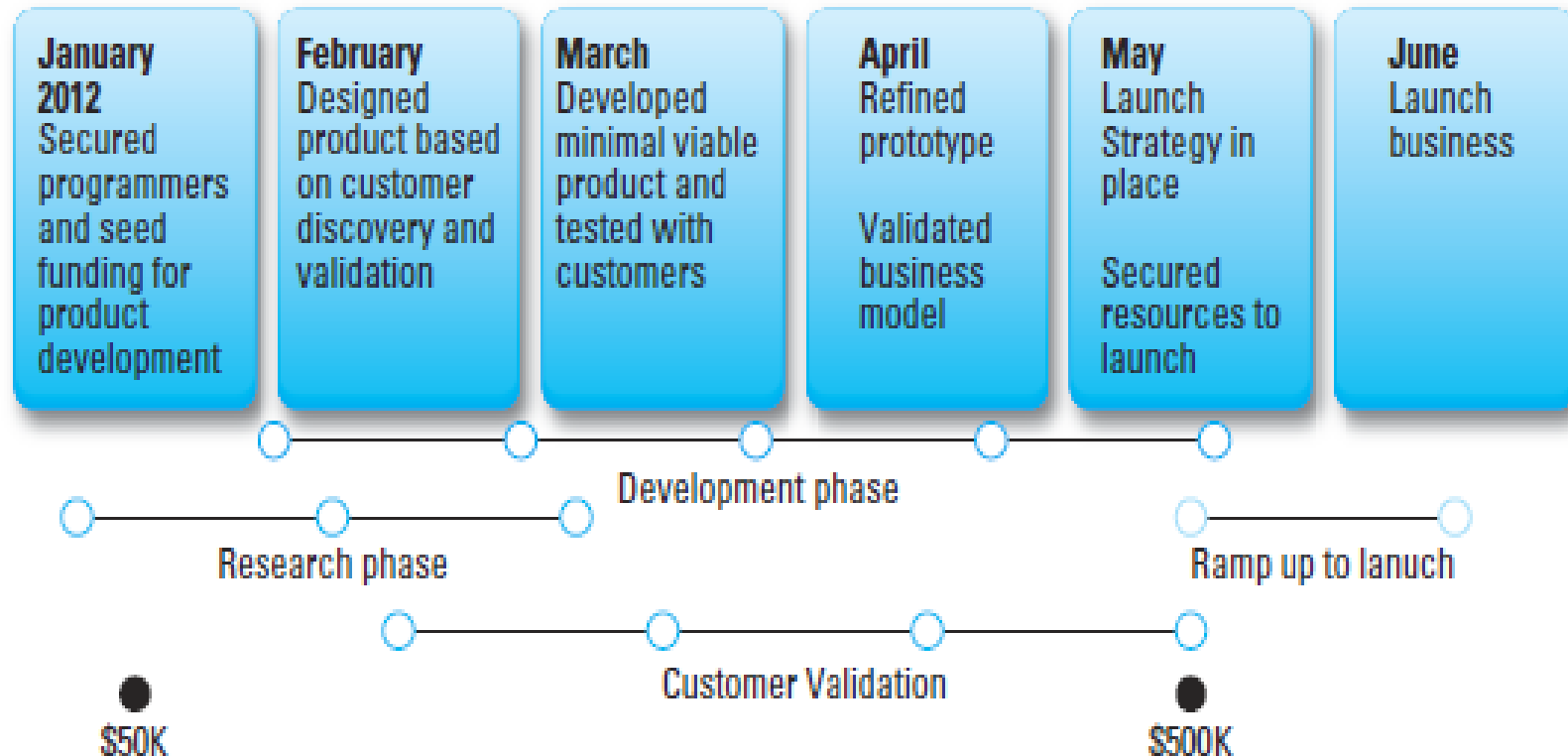
- ❑ Where your new venture lies in the value chain will determine what its margins are, who its customer is, and how much it can charge for its products and services
- ❑ A service business delivers services direct to a customer, but product businesses are different
 - Is the venture a manufacturer or producer, wholesaler or distributor, or retailer dealing with consumers?
 - Each produces different margins, prices, logistics

9.1c Create a Timeline & Milestones for Resource Requirements

- Note that Figure 9.3 depicts a timeline to launch for a hypothetical product
 - It is presented on a very aggressive timeline that is not typical. At each monthly milestone, the team expects to complete a specific goal
 - Also, initial funding requirements have been identified and associated with a milestone
 - The timeline should extend through the first year, as there will be ups and downs

Figure 9.3- Timeline and Milestones to Launch

Project Start Date



9.1d Define Your Launch Strategy to Reduce Risk

- ❑ You want to identify as many potential risks as possible and eliminate the right ones
- ❑ Risks are “known, unknowns, and unknown unknowns”
 - Deal-killer risks
 - Prevent the business from launching
 - Path-dependent risks
 - Arise from taking the wrong path
 - Low-hanging fruit with high ROI
 - These are easy and cheap to resolve; solve them

9.2 Startup Financial Metrics

❑ Some common metrics all startups use are:

- Sales forecast
- Headcount
- Expenses (fixed and variable)
- Break-even cash flow

❑ Also, specific types of businesses have unique metrics; some are:

- Gross margin
 - Inventory turns
 - Occupancy
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9.2 Startup Financial Metrics

Some specific metrics (cont.):

- Qualified leads
- Customer acquisition costs (CAC)
- Average order size, time to reorder & lifetime value per customer
- Revenues per salesperson and time to revenue for direct sales
- For Internet ventures: acquisition, retention, revenue, viral coefficient
 - Table 9.1 presents some Internet metrics

Table 9.1- Metrics for Web 2.0 Ventures

TABLE 9.1 Metrics for Web 2.0 Ventures

	Visitor	Contributor	Distributor
	Total unique visitors Total page views Total visits		Shares content with others
Acquisition (Registrations and activations)	Watch a video	Submit a video or other content	Shares a video with other users and other sites
Retention	Visit once a week for 3 months	Submits content once a month	Shares video that drives X visits a month for 3 months
Revenue	Click on an ad (qualified lead)	Pays for subscription or premium membership	Drives X number of premium users

9.3 Develop Financial Assumptions

The most important part of any analysis of financial feasibility is the assumptions on which the analysis is based; one example:

Customer acquisition costs—The company will sell via a direct sales model, which requires regular client contact. In addition, the product plan calls for launching both the core product and the ancillaries at well-attended industry conferences to maximize the value of the presentations. We estimate the associated costs to be as follows:

1. Customer site visits: 3–4 trips per month beginning July 7 @ \$800 each for travel expenses.
2. Booth price for Gartner-sponsored data mining conference: \$3,000 as provided by Gartner representative.
3. Travel to/from conference for two attendees: 3 conferences, 2 people @ \$2,200 each to include transportation, hotel, and meal allowance.⁹

9.3a Estimate New Product/Service Demand

❏ Must triangulate demand from 3 different viewpoints:

- 1. historical analogy with similar products
- 2. customer feedback, end-user and intermediary feedback
- 3. the entrepreneur's own perspective, gleaned from previous experience, and going into limited production or doing a test market

9.3a Estimate New Product/Service Demand

To estimate demand:

- Use historical analogy or substitute products
- Talk to customers and intermediaries
- Use your knowledge and experience
- Go into limited production

9.3b Pricing Assumptions

- ❑ Entrepreneurs typically have to price their products long before they know the exact costs of producing it, and before they have a good handle on the price a customer will pay
- ❑ Pricing becomes the central selling point when the product or service is a commodity, when the only feature differentiating the product or service from those offered by competitors is price

9.3b Pricing Assumptions

- ❑ Entrepreneurs typically set prices based on cost to produce plus a margin
- ❑ But fixed costs should not be apportioned within the price because they do not come into play when generating additional sales
 - It is the contribution margin that affects profitability'
- ❑ Every industry has discounts, including cash and quantity discounts
 - These should be factored in in calculating price

9.3b Pricing Assumptions

- ❑ Other factors to consider in addition to cost are based in part on a company's goals
 - If the goal is to increase sales or market share, prices may need to be lowered to increase the volume sold
 - If the goal is to maximize cash flow, raising prices and reducing direct costs and overhead may be the answer
 - If the goal is to define an image, raise the price
 - To control demand in scarcity, raise prices

Figure 9.4- Customer Goals and Price

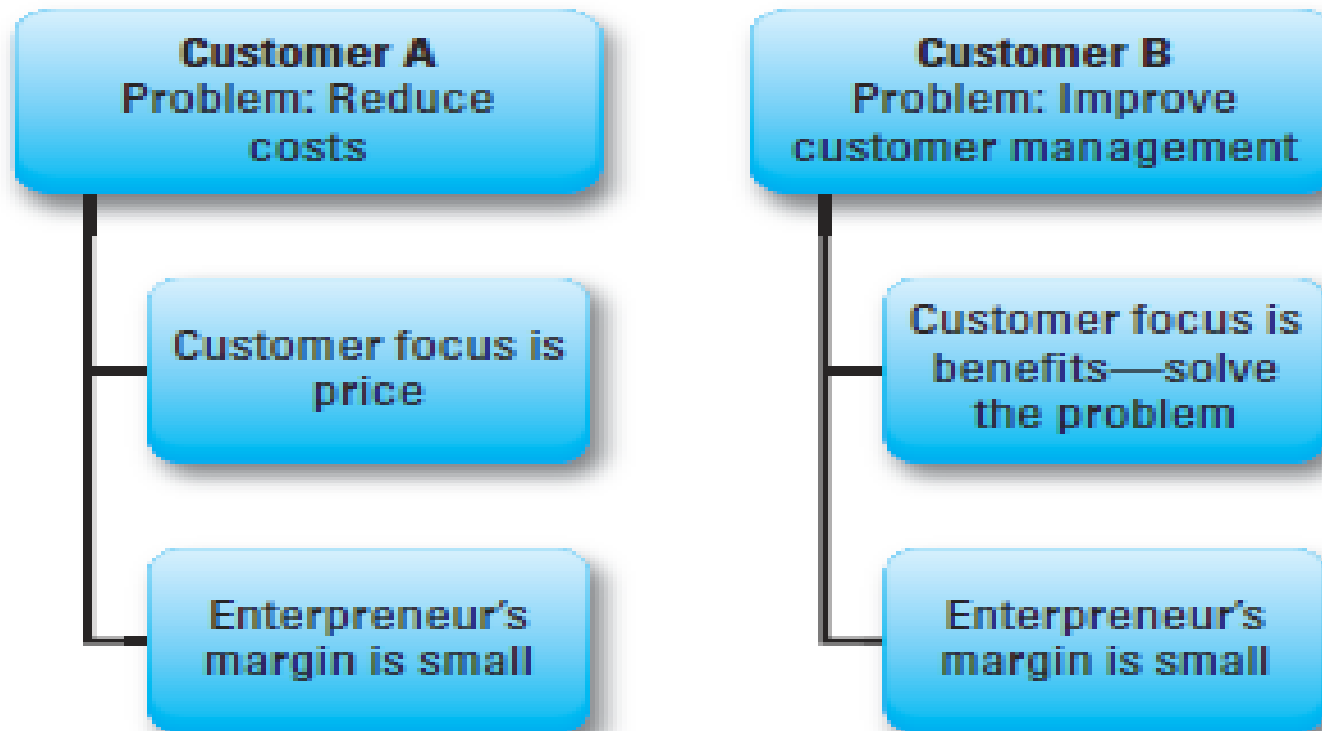


Table 9.2- Common Pricing Strategies at Startup

Premium Pricing	Uses a high price to reflect a unique product/service and a significant competitive advantage.
Price Skimming	Starts with a high price to capture uniqueness and competitive advantage. Then as new competitors enter the market, drops the price to stay ahead of competition.
Demand-Based Pricing	Find out what customers are willing to pay for the product and price it accordingly.
Captive Product Pricing	Where the entrepreneur's product has complements, charge a low price for source product (i.e., a printer) and a premium for consumables (i.e., ink cartridges).
Psychological Pricing	To create a complex pricing structure by combining multiple products and services into one package.
Product Bundle Pricing	In a channel with many intermediaries (distributors, retailers), the final price to the consumer or end-user must be tolerable, given all the markups along the value chain. Compare what the market will bear with the cost of getting a product to market.
Geographical Pricing	Used where there are price variations in different geographical locations where the product is sold.

9.3b Pricing Assumptions

- ❑ There are no formulas that will give your new venture with no track record a solution to the problem of pricing, and firms won't discuss it
 - ❑ But by triangulating with costs, competitor pricing and feedback from customers and value chain partners, you can reach a number that can be tested in the market
 - ❑ Most important to understand customer behavior
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9.3c Estimate Revenues, Expenses and Startup Costs

1. For manufactured products, get a physical prototype early to calculate costs
2. Product development may take months to years, and costs for prototyping are higher than ultimate production costs; need a prototype to determine economic feasibility
3. For service firms, costs must be based on competitors' experience
4. Entrepreneurs often start ventures in industries they know, to gain information

9.3c Estimate Revenues, Expenses and Startup Costs

- ❑ *Sales Forecast* should be calculated first because the volume of sales affects expenses
- ❑ *Expenses* forecasting for manufacturing complex because COGS must be derived first
- ❑ *Startup Costs* are mostly incurred before the business opens its doors (furniture, supplies)
- ❑ *Keeping the Numbers Real* is important because they will be challenged and must be defended

Table 9.3- Redfin: How Startup Expenses Estimates Change

Type of Expense	Redfin's Model	Actual Expenditures
Rent per employee per month	\$250	\$336
Initial per-employee equipment cost	\$6,500	\$5,700
Annual payroll tax	12.5%	8.5%
Monthly travel costs	\$300	\$369
Monthly telephone costs per field employee	\$125	\$261
Annual accounting costs	\$45,000	\$32,912

Source: Kelman, G. (October 1, 2007). "Financial Models for Underachievers: Two Years of the Real Numbers of a Startup," *How to Change the World*, <http://blog.guykawasaki.com/2007/10/financial-model.html#axzzOuM3rRMgX>.

9.4 Calculating a Startup's Cash Requirements

- ❑ First, develop cash flow statement (Fig 9.5)
- ❑ Because this is an estimate based on a series of estimates, there is a good chance that it is not entirely accurate, so entrepreneurs typically add a safety margin
- ❑ Table 9.4 presents a breakout of the startup capital requirements for a mobile payment company
- ❑ Note that cash needs are separated into types of capital resources required

Figure 9.5- Sample Direct Cash Flow Statement...

Cash Outflows	Mo.1	2	3	4	5	6	7	8	9	10	11	12	13	14
Office Space w/ Utilities	0	0	0	0	0	0	0	0	0	0	0	0	500	500
Office Equipment	50	50	50	50	50	50	50	50	50	50	50	50	750	750
Advertising	750	750	750	750	750	750	750	750	750	750	750	750	1,000	1,000
Travel & Entertainment	1,000	1,000	1,000	1,000	1,000	1,000	2,000	2,000	2,000	2,000	2,000	2,000	2,000	2,000
Accounting	1,000	1,000	1,000	1,000	1,000	1,000	1,000	1,000	1,000	1,000	1,000	1,000	1,000	1,000
Legal	1,500	1,500	1,500	1,500	1,500	1,500	1,500	1,500	1,500	1,500	1,500	1,500	1,500	1,500
Insurance	1,000	1,000	1,000	1,000	1,000	1,000	1,000	1,000	1,000	1,000	1,000	1,000	1,000	1,000
FTE Load	3,500	3,500	3,500	3,500	3,500	3,500	3,500	3,500	3,500	3,500	3,500	3,500	7,000	7,000
Total Fixed Costs	324,250	19,250	19,250	19,250	19,250	23,250	21,000	25,000	21,743	34,993	29,235	30,235	148,728	47,978
Variable Costs														
Server Lease	1,200	1,200	1,200	1,200	1,200	2,700	2,700	4,200	4,200	5,700	5,700	7,200	8,700	8,700
Transportation	1,500	1,500	1,500	1,500	1,500	3,000	3,000	4,500	4,500	6,000	6,000	7,500	9,000	9,000
Installation Costs	500	500	500	500	500	1,000	500	1,000	500	1,000	500	1,000	1,000	500
Cell Phones	500	500	500	500	500	500	500	500	500	750	750	750	1,000	1,000
Internet	500	500	500	500	500	500	500	500	500	750	750	750	1,000	1,000
Total Variable Costs	4,200	4,200	4,200	4,200	4,200	7,700	7,200	10,700	10,200	14,200	13,200	17,200	20,700	20,200
Total Cash Outflows	328,450	23,450	23,450	23,450	23,450	30,950	28,200	35,700	31,943	49,193	42,935	50,435	169,428	63,178
Net Cash In/Outflow	-328,450	-23,500	-23,335	-23,319	-23,301	9,220	-20,506	12,021	-16,765	6,020	-20,357	12,289	-99,226	-25,416
Cumulative Cash Flow	-328,450	-351,950	-375,135	-398,454	-421,754	-412,534	-433,040	-421,018	-437,783	-431,763	-452,020	-439,731	-538,957	-564,374

Table 9.4- Sample Cash Needs Table...

Capital Expenditures (CE)	In Dollars	
Equipment	50,000	
Pre-Operating Startup Costs (POSU)		
R&D Costs	250,000	
Salaries	10,000	
FTE Workstations	5,000	
Office Equipment	500	
Advertising	750	
Travel & entertainment	1,000	
Accounting	1,000	
Legal	1,000	
Insurance	1,000	
FTE Load	3,500	
Server Lease	1,200	
Transportation	1,500	
Installation Costs	500	
Cell Phones	500	
Internet	500	
Total Pre-Op SU	277,950	
Working Capital (WC)		
Working Capital Needs	236,424	Difference between highest cumulative negative CF and CAPEX + Pre-Op SU
Highest Cumulative Negative CF	564,374	
Safety Factor (SF)		
Safety Factor	77,000	120 days of fixed expenses
Total Startup Capital	641,374	

9.4a Assessing Risk

- ❑ What would be the effect on the financials of a change in price? A decline in sales? Or unexpected demand? Ask other questions
- ❑ Consider potential changes most likely to happen; ask
 - Do startup capital requirements make sense?
 - Looking at the capital investment and profit possibilities, is there enough money here to make the effort worthwhile?
- ❑ Keep checking assumptions

New Venture Action Plan

- ☐ Determine the startup metrics for your company
 - ☐ Gather the numbers you need for performing your financial analysis
 - ☐ Gather sales forecast data through triangulation
 - ☐ Create a cash flow statement from startup until a positive cash flow is achieved
 - ☐ Perform a cash requirements assessment to determine how much capital you will need to start the business
 - ☐ Determine whether this venture is financially feasible
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